Karmayogi Institute Of Technology ,

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Department of Computer Technology

Micro-Project Report

Programming With Python-22616

" Various Methods or Functions in Python…"

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❖Aims/Benefits of the Micro-Project:

Using methods in programming, such as in languages like Java, Python, or C++, offers several advantages: 1. **Modularity**: Methods allow you to break down a complex program into smaller, more manageable pieces. Each method can perform a specific task, making your code easier to understand and maintain

❖Course Outcomes Achieved:

- ❖ Create your first program in Python IDLE.
- ❖ Implement OOPs concepts in your programming.
- ❖ Use Arrays, and Data structures.
- ❖ Create an application with the support of graphics in Python.
- ❖ Implement error handling.

Hardware computer system:

Computer (i5-i3 preferable), RAM minimum 2GB and onwards, HDD of 160 GB

System Operating

Windows 7 And LINUX version 5.0 or later.

Software

Any software.

❖Action Methodology Followed:

- **del: This is a keyword in Python used to delete elements from a list or dictionary by index or key.**
- **clear(): This method is used to remove all elements from a list.**
- **remove(): This method removes the first occurrence of a specified value from a list.**
- **pop(): This method removes the element at the specified position (index) from the list and returns it.**
- **append(): This method adds an element to the end of the list.**
- **insert(): This method inserts an element at the specified position (index) in the list.**
- **extend(): This method adds all the elements of an iterable (such as a list) to the end of the list.**

- **min(): This function returns the smallest element in a list.**

- **max(): This function returns the largest element in a list.**
- **len(): This function returns the number of elements in an object (such as a list, tuple, or set).**
- **add(): This method adds an element to a set. If the element is already present, it doesn't add it again.**
- **reverse(): This method reverses the elements of a list in place.**

  sort(): This method sorts the elements of a list in ascending order. It can take parameters to customize sorting behavior.

Program Code:

```
print("******Functions in Python...********\n\n")
l=[10,20,30,40,50,60,"SD",70,80,90]
list=[2,3,4,5,6]
d=[2,3,4,5,6,7]
t1=(10,20,"SD",99,70.48,96)
t2=(40,60,"SD",98,79.48,86)
s={'Sujata','Dipali','Amruta','Sanika','Shreya',"ASD"}


print("Original List..\n",l)
print("\nOriginal First Tuple..\n",t1)
print("\nOriginal Second Tuple..\n",t2)
print("\nOriginal Set..\n",s)


print("\nVarious operations perform on List,Tuple and Set..\n")


print("\n1.Using del keyword..")
print(l)
del(l[2])
print("After deleting element\n",l)


print("\n2.Using clear function...")
print(list)
list.clear()
print("After clear element\n",list)
```

```
print("\n3.Using remove function...")
print(l)
l.remove(10)
print("After removing element\n",l)

print("\n4.Using Pop function...")
print(l)
l.pop(5)
print("After using pop function\n",l)

print("\n5.Using append function...")
print(l)
l.append(92)
print("After using append function\n",l)

print("\n6.Using insert function...")
print(l)
l.insert(7,89)
print("After using insert function\n",l)

print("\n7.Using extend function...")
print(l)
l.extend([88,"FAB5","ASD",99])
print("After using extend function\n",l)

print("\n8.Using min function...")
print(d)

print("After using min function\n",min(d))
```

```
print("\n9.Using max function...")

print(d)

print("After using max function\n",max(d))


print("\n10.Using length function...")

print(s)

print("After using length function\n",len(s))



print("\n11.Using add method...")

print(s)

s.add("FAB5")

print("After using add method\n",s)


print("\n12.Using Reverse method...")

print(d)

d.reverse

print("After using Reverse method\n\n",d)


print("\n13.Using sort method...")

print(d)

d.sort()

print("After using sort method\n\n",d)
```

❖ Actual Required Followed:

N o.	Name of Resource/material	Specifications	Quantity	Remarks
1	**Hardware computer system**	Computer (i5-i3 preferable),RAM minimum 2GB and onwards,HDD of 160 GB	-	
2	**Operating system**	Windows 7 And LINUX version 5.0 or later	-	
3	**Software**	IDLE	-	

❖ Outputs of the Micro-Projects:

********Functions in Python...**********

Original List..
[10, 20, 30, 40, 50, 60, 'SD', 70, 80, 90]

Original First Tuple..
(10, 20, 'SD', 99, 70.48, 96)

Original Second Tuple..
(40, 60, 'SD', 98, 79.48, 86)

Original Set..
{'Sujata', 'Sanika', 'Amruta', 'Shreya', 'ASD', 'Dipali'}

Various operations perform on List,Tuple and Set..

1.Using del keyword..
[10, 20, 30, 40, 50, 60, 'SD', 70, 80, 90]
After deleting element
[10, 20, 40, 50, 60, 'SD', 70, 80, 90]

2.Using clear function...
[2, 3, 4, 5, 6]
After clear element
[]

3.Using remove function...
[10, 20, 40, 50, 60, 'SD', 70, 80, 90]
After removing element
[20, 40, 50, 60, 'SD', 70, 80, 90]

4.Using Pop function...
[20, 40, 50, 60, 'SD', 70, 80, 90]
After using pop function
[20, 40, 50, 60, 'SD', 80, 90]

5.Using append function...
[20, 40, 50, 60, 'SD', 80, 90]
After using append function
[20, 40, 50, 60, 'SD', 80, 90, 92]

6.Using insert function...
[20, 40, 50, 60, 'SD', 80, 90, 92]
After using insert function
[20, 40, 50, 60, 'SD', 80, 90, 89, 92]

7.Using extend function...
[20, 40, 50, 60, 'SD', 80, 90, 89, 92]
After using extend function
[20, 40, 50, 60, 'SD', 80, 90, 89, 92, 88, 'FAB5', 'ASD', 99]

8.Using min function...

[2, 3, 4, 5, 6, 7]
After using min function
2

9.Using max function...
[2, 3, 4, 5, 6, 7]
After using max function
7

10.Using length function...
{'Sujata', 'Sanika', 'Amruta', 'Shreya', 'ASD', 'Dipali'}
After using length function
6

11.Using add method...
{'Sujata', 'Sanika', 'Amruta', 'Shreya', 'ASD', 'Dipali'}
After using add method
{'FAB5', 'Sujata', 'Sanika', 'Amruta', 'Shreya', 'ASD', 'Dipali'}

12.Using Reverse method...
[2, 3, 4, 5, 6, 7]
After using Reverse method

[2, 3, 4, 5, 6, 7]

13.Using sort method...
[2, 3, 4, 5, 6, 7]
After using sort method

[2, 3, 4, 5, 6, 7]

❖ Applications:

- Web Development.
- Data Science — including machine learning, data analysis, and data visualization.
- Scripting

❖ Advantages :

In Python, methods are functions that are associated with an object and can manipulate its data or perform actions on it. They are called using dot notation, with the object name followed by a period and the method name. Methods are an important part of object-oriented programming in Python

❖ Disadvantages:

- ❖ Slow Speed. Python is a dynamically typed and interpreted language, as previously mentioned. ...
- ❖ Not Memory Efficient. Python must make a small tradeoff to make development simpler. ...
- ❖ Weak Mobile Computing. Most of the time, server-side programming calls for Python. ...
- ❖ Database Access. ...
- ❖ Runtime Errors.

❖ Conclusion :

Extended classes and methods in Python provide a powerful way to create flexible and modular code. By understanding and using inheritance effectively, you can streamline your programming tasks and make your code more organized and maintainable

❖ Skill Developed by this Project:

Although python is easy to learn, it can sometimes present difficulties, especially if you are learning the complex aspects of the language.
For example, you can find motivation in the fact that you're likely to get a promotion in your current role or get your dream job once you acquire advanced Python skills.

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